

**Laxmi Charitable Trust's**  
**Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce**  
**Department of Information Technology (B.Sc.I.T Semester V)**  
**Dot .Net Core and Progressive Web Application Practical**

**Practical – 3**

|                                |                                     |
|--------------------------------|-------------------------------------|
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| Class: TYIT                    | Batch: 1                            |
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**Part B - Program**

a) Create methods add(), multiply(), subtract(), divide() with suitable parameters and call these methods using concept of C# delegate.

**Algorithm:**

Step1 :-Start the program.

step2:-Create a delegate and define calculator methods (Add, Sub, Mul, Div).

step3:-Create an object of the Calculator class.

step4:-Assign each method to the delegate and invoke it.

Step5:- Display the results and stop the program.

**Code:**

```
using System;
delegate void operation(int a, int b);
class Calculator
{
    public void Add(int a, int b)
    {
        Console.WriteLine("Addition =" + (a + b));
    }
    public void Sub(int a, int b)
    {
        Console.WriteLine("Subtraction =" + (a - b));
    }
    public void Mul(int a, int b)
    {
        Console.WriteLine("Multiplication =" + (a * b));
    }
    public void Div(int a, int b)
    {
        Console.WriteLine("Division =" + (a / b));
    }
}
class program
{
    static void Main()
    {
        Calculator c = new Calculator();
        operation obj;
```

```
obj = c.Add;
obj(10,10);
obj = c.Sub;
obj(20, 10);
obj = c.Mul;
obj(20, 10);
obj = c.Div;
obj(8, 4);
Console.WriteLine("Aashu 50");
}
}
```

### Output:

```
Addition =20
Subtraction =10
Multiplication =200
Division =2
Aashu 50
```

b) Write a program using multicast delegate.

### Algorithm:

step1:-Start the program.

step2:-Create a delegate and define the methods Welcome(), Learn(), and Thanks().

step3:-Create an object of the Demo class.

step4:-Add all methods to the delegate and invoke it.

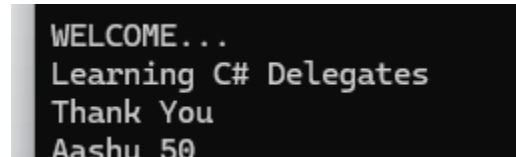
step5:-Display the messages and stop the program.

### Code:

```
using System;
delegate void Message();
class Demo
{
    public void Welcome()
    {
        Console.WriteLine("WELCOME...");
    }
    public void Learn()
    {
        Console.WriteLine("Learning C# Delegates");
    }
    public void Thanks()
    {
        Console.WriteLine("Thank You");
    }
}
public class Program
{
    public static void Main(string[] args)
    {
        Demo d = new Demo();
```

```
        Message m;  
        m = d.Welcome;  
        m += d.Learn;  
        m += d.Thanks;  
        m();  
        Console.WriteLine("Aashu 50");  
    }  
}
```

**Output:**

A screenshot of a console window with a black background and white text. The text is displayed in four lines: "WELCOME...", "Learning C# Delegates", "Thank You", and "Aashu 50".

```
WELCOME...  
Learning C# Delegates  
Thank You  
Aashu 50
```

c) Create a class BankAccount with AccountNumber and Balance. Implement property for Balance with validation (Balance cannot be negative).

**Algorithm:**

step1:-Start the program.

step2:-Create the BankAccount class with AccountNumber and Balance properties.

step3:-Create an object of the BankAccount class.

step4:-Assign and display the account number and balance.

step5:-Update the balance, display the details, and stop the program.

**Code:**

```
using System;  
class BankAccount  
{  
    public int AccountNumber { get; set; }  
    private double balance;  
    public double Balance  
    {  
        get  
        {  
            return balance;  
        }  
        set  
        {  
            if (balance >= 0)  
                balance = value;  
            else  
                Console.WriteLine("Balance cannot be negative");  
        }  
    }  
}  
class program  
{  
    static void Main()  
    {  
        BankAccount b = new BankAccount();  
    }  
}
```

```
b.AccountNumber = 110;  
b.Balance = 5000;  
Console.WriteLine("Account Number " + b.AccountNumber);  
Console.WriteLine("Balance" + b.Balance);  
b.Balance = 1000;  
Console.WriteLine("Aashu 50");  
}  
}  
}
```

**Output:**

```
Account number:101  
Balance:5000  
aashu 50
```